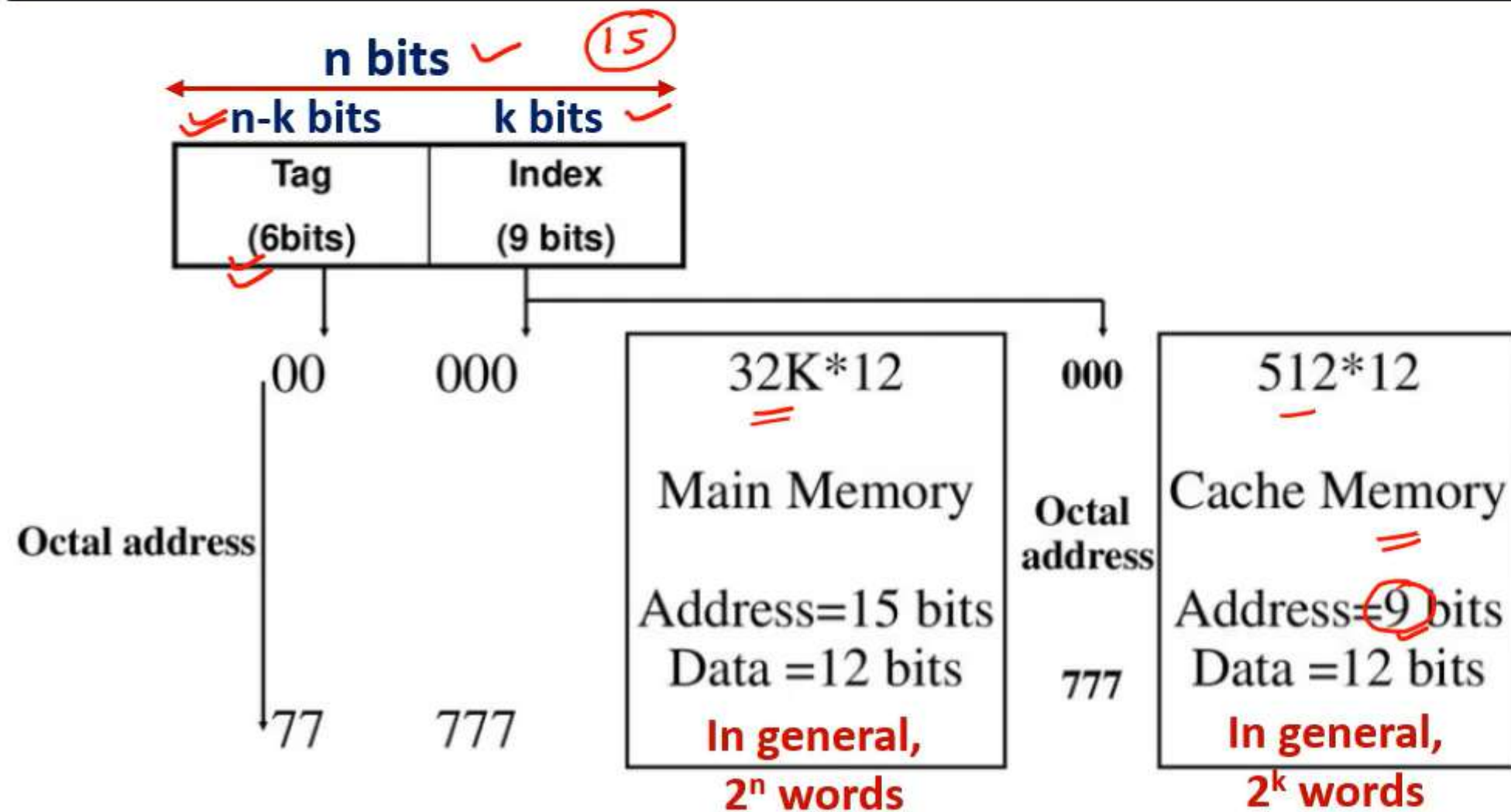


Direct Mapping Cache

- ✓ ☐ A direct-mapped cache is the **simplest approach**, where each main memory address maps to **exactly one** cache block
- ✓ ☐ Main memory needs an **address** that **includes** both the tag and the index bits
- ☐ **The number of bits** in the index field is equal to the number of address bits required to access the cache memory
- ☐ **The direct mapping cache** organization uses the n-bits address to access the main memory and the k-bits index to access the cache



- **Main memory** can store 32K words of 12 bits each
- **The cache** is capable of storing 512 of these words
- For every word stored in cache, there is a **duplicate** copy in main memory
- **The CPU** communicates with both memories, it first sends a 15-bit address to cache
- If there is a **hit**, the CPU accepts the 12-bit data from cache
- ✓ ▪ If there is a **miss**, the CPU reads the word from main memory and the word is then transferred to cache

Direct Mapping Cache

A particular block of main memory can map to only one particular line of the cache

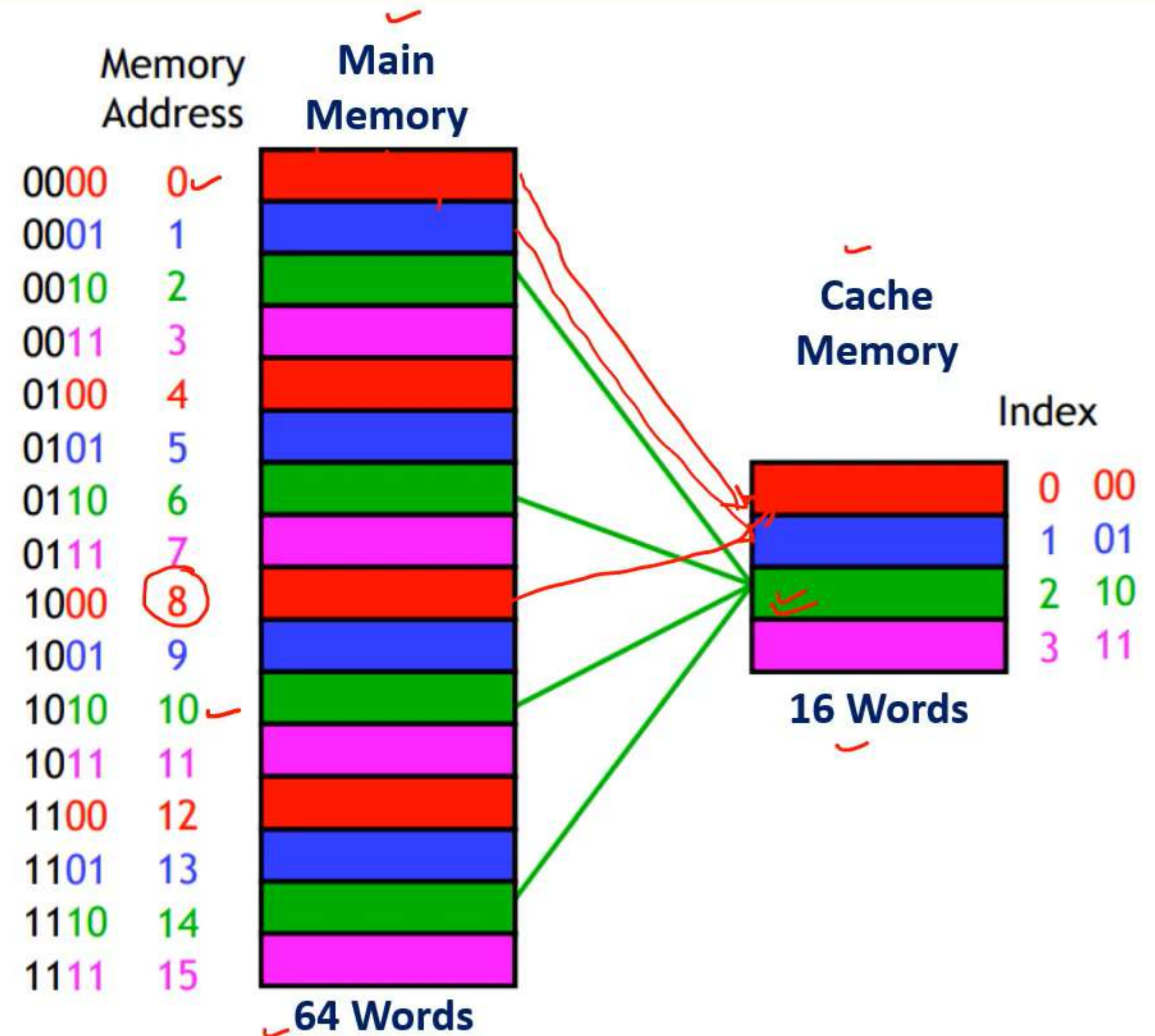
✓ Cache line number = (Main Memory Block Address)
Modulo (Number of lines in Cache)

mod operator i.e. (remainder)

$0 \bmod 4 = 0$
 $1 \bmod 4 = 1$
 $8 \bmod 4 = 0$
 $10 \bmod 4 = 2$

- ❑ Example: there is a 16-byte main memory and a 4-byte cache
- ✓❑ Memory locations 0, 4, 8 and 12 all map to cache block 0
- ❑ Addresses 1, 5, 9 and 13 map to cache block 1, etc.
- ❑ For instance, with the four-block cache here, address 14 would map to cache block 2

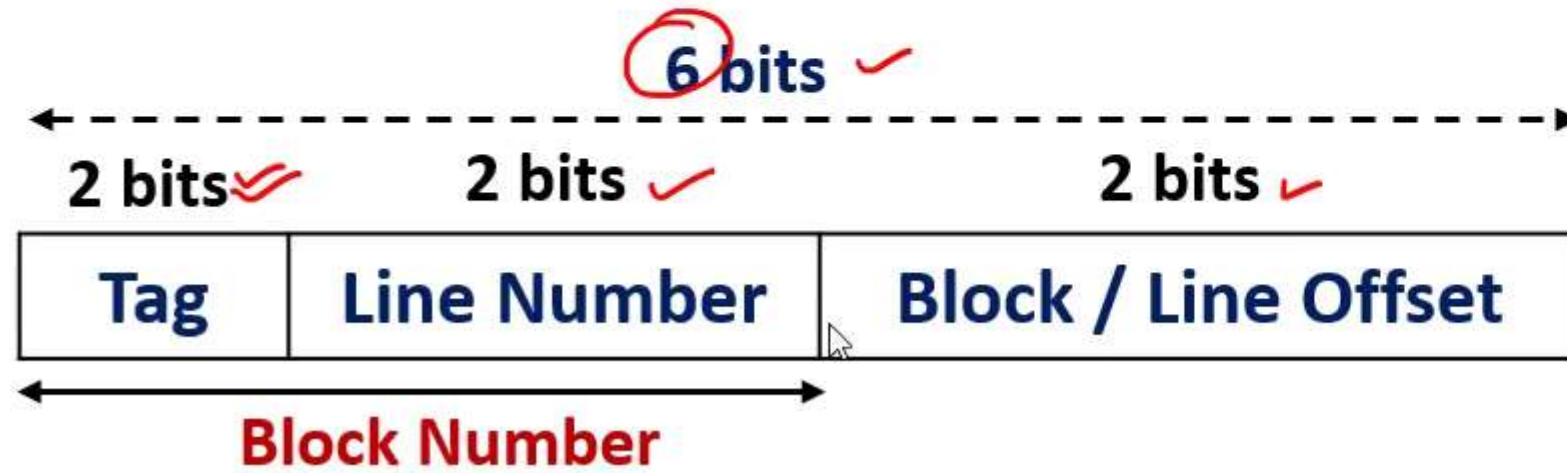
$$14 \bmod 4 = 2$$



The Direct Mapping Cache Organization

Direct Mapping Cache

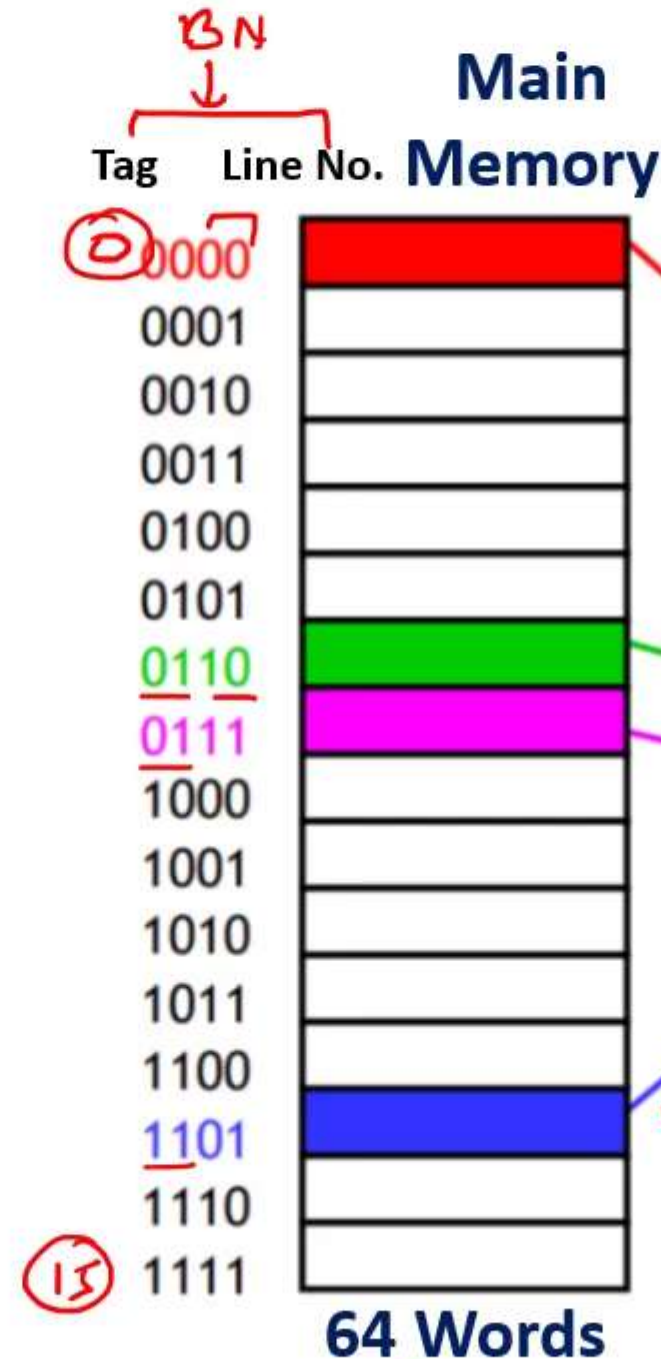
Division of Physical Address



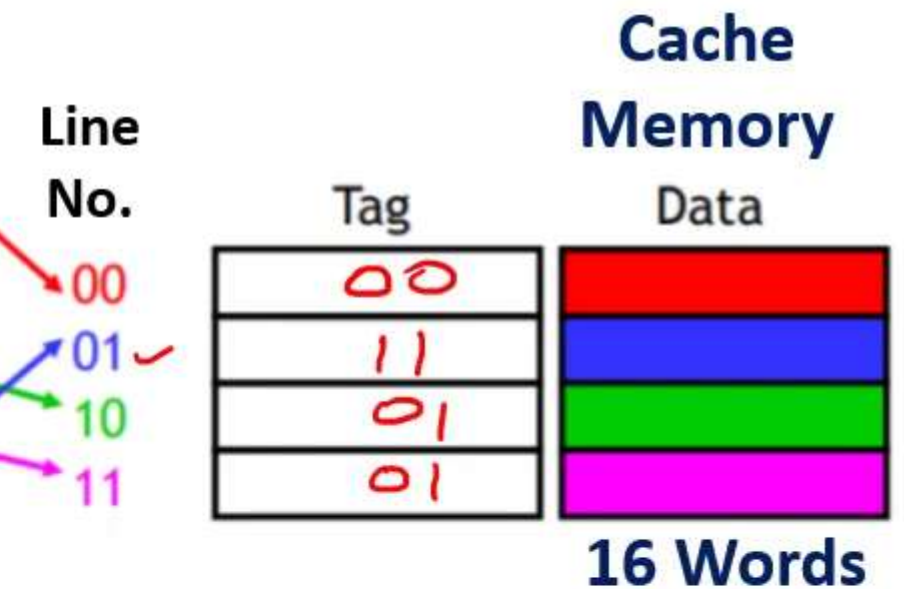
Total words: 64 → 2^6
 Block Size: 4 → 2^2
 Total Line No's: 4 → 2^2
 Total Blocks: 16 → 2^4 BN

Example: Physical Address - 00 00 01

Tag	Line Number	Block Offset
-----	-------------	--------------

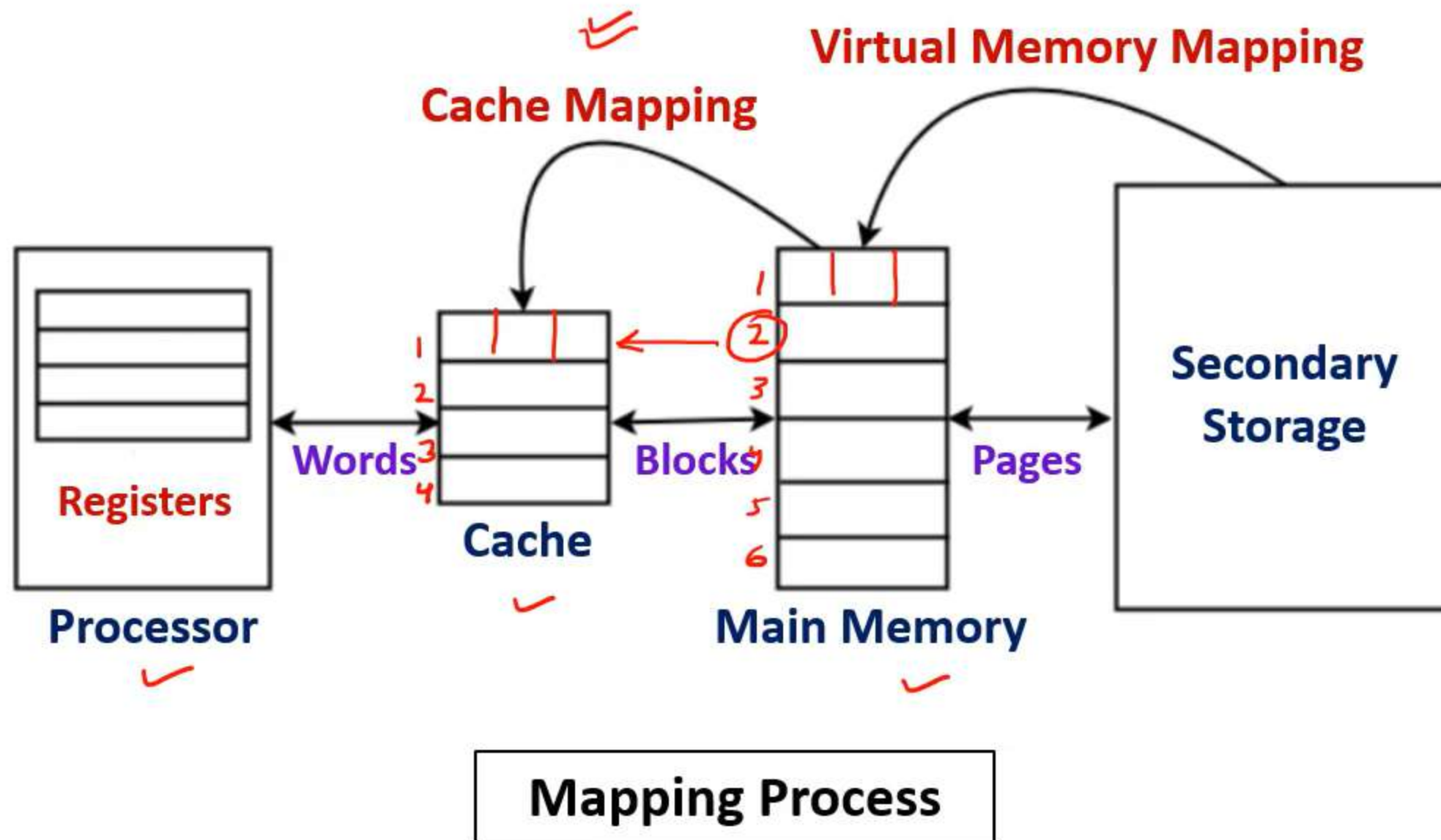


Block No.	Tag	Line No.
0000	00	00
0001	00	01
0010	00	10



Cache Memory Mapping

- ❑ It **defines** how a block from the main memory is mapped to the cache memory in case of a cache miss
- OR
- ✓❑ It is a **technique** using which the contents of main memory are brought into the cache memory



- **Main memory** is divided into equal size partitions known as **blocks** or **frames**
- **Cache Memory** is divided into partitions having same size as that of blocks known as **lines**
- During cache mapping, block of main memory is simply **copied** to the cache and the block is **not** actually **brought** at all from the main memory

Cache Memory Mapping

Cache Mapping Techniques

Direct Mapping

- ❖ The simplest technique
- ❖ It maps each block of main memory into only one possible cache line
- ❖ Less expensive than associative mapping
- ❖ Fast as only tag field matching is required while searching for a word

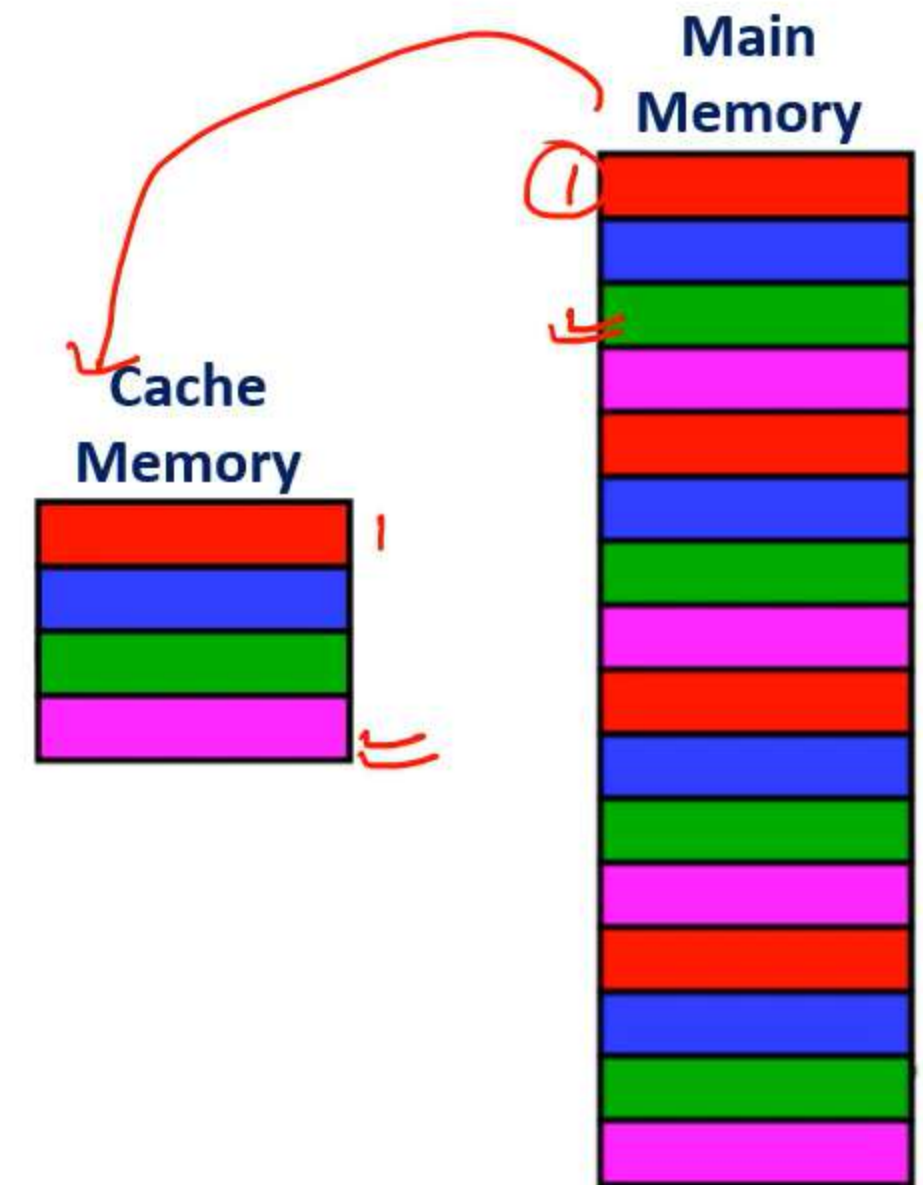
Associative Mapping

- ❖ A block of main memory can map to any line of the cache that is freely available at that moment
- ❖ It is fast
- ❖ Easy to implement

Set-associative Mapping

- ❖ Cache lines are grouped into sets
- ❖ A particular block of main memory can map to only one particular set of the cache
- ❖ Gives better performance than other two mapping techniques

Cache mapping refers to a technique using which we bring the main memory into the cache memory



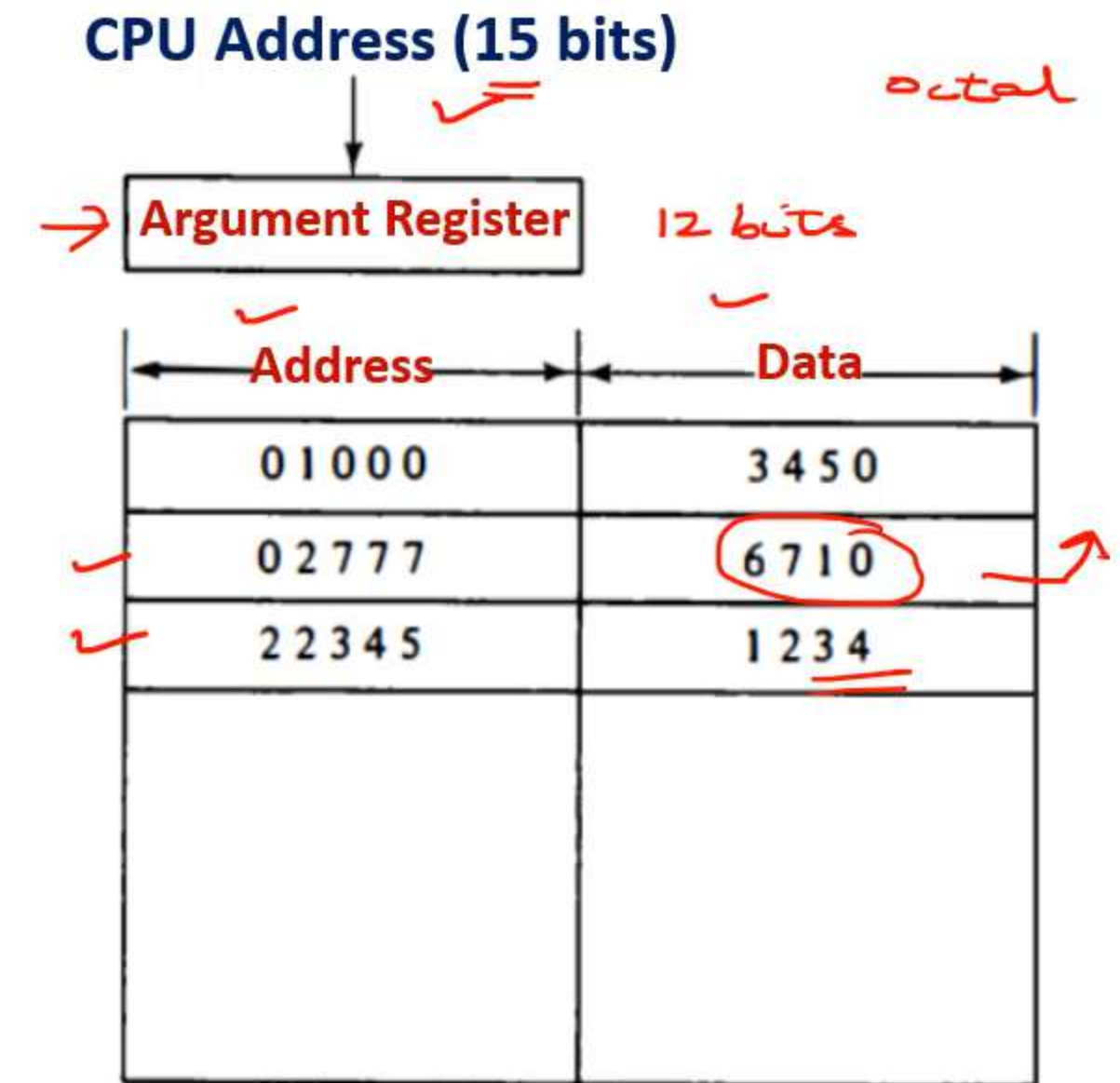
Associative Mapping

- ❑ The **fastest** and **most flexible** Cache organization uses an Associative Memory
- ❑ A **block of main memory** can be **mapped** to **any freely available cache line**, this makes fully associative mapping more flexible than direct mapping

- If **cache** is **full** then a **decision** must be taken to which line in cache to be **replaced** by the new content (i.e. need a replacement algorithm)
- A **replacement algorithm** is needed to replace a block if the cache is full

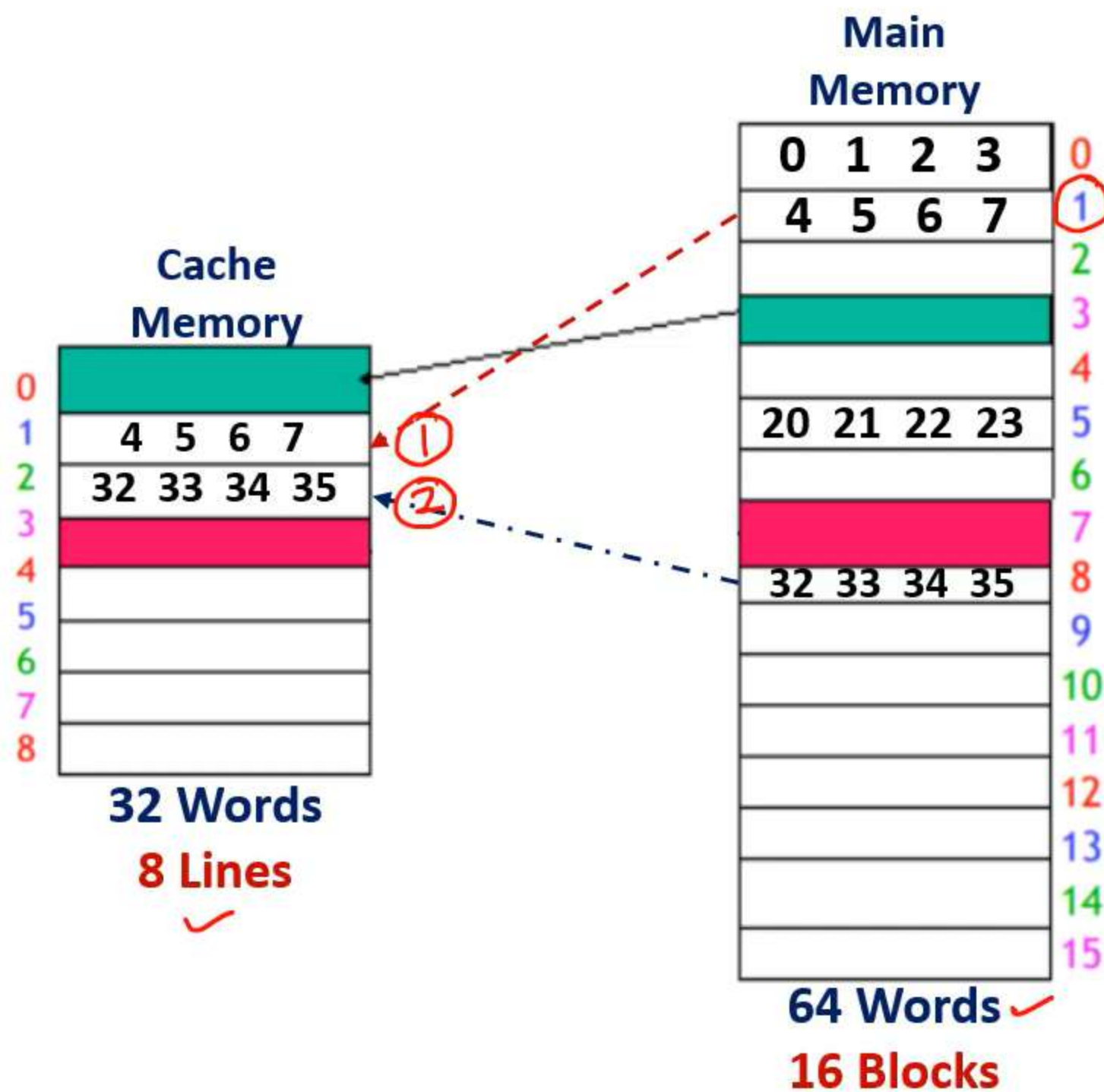
Few popular replacement Algorithms are:

- ❖ **Least Recently Used (LRU)**: This approach restores the block that has not been read by the processor in the highest period
- ❖ **First In First Out (FIFO)**: This approach restores the block that has been in cache the highest
- ❖ **Least Frequently Used (LFU)**: This approach restores the block which has fewer hits because being loaded into the cache
- ❖ **Random Replacement**: This approach randomly chooses a block to be replaced. Its execution is slightly lower than LRU, FIFO or LFU.

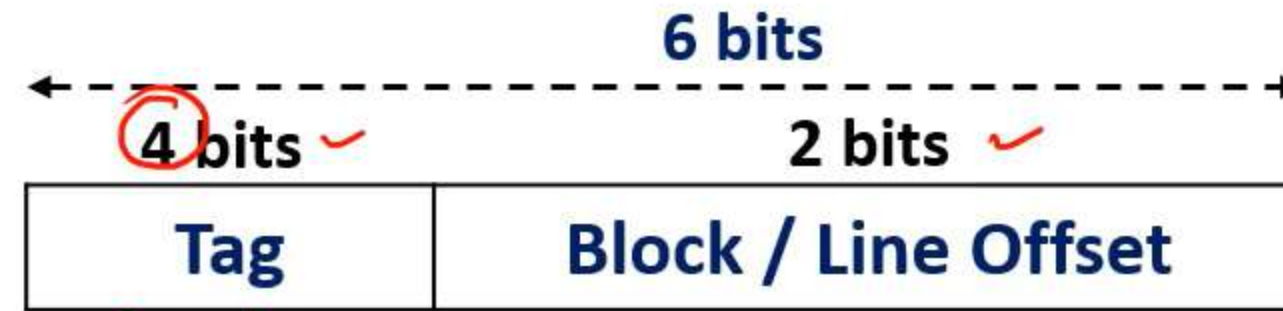


Associative Mapping Cache (all numbers in Octal)

Associative Mapping



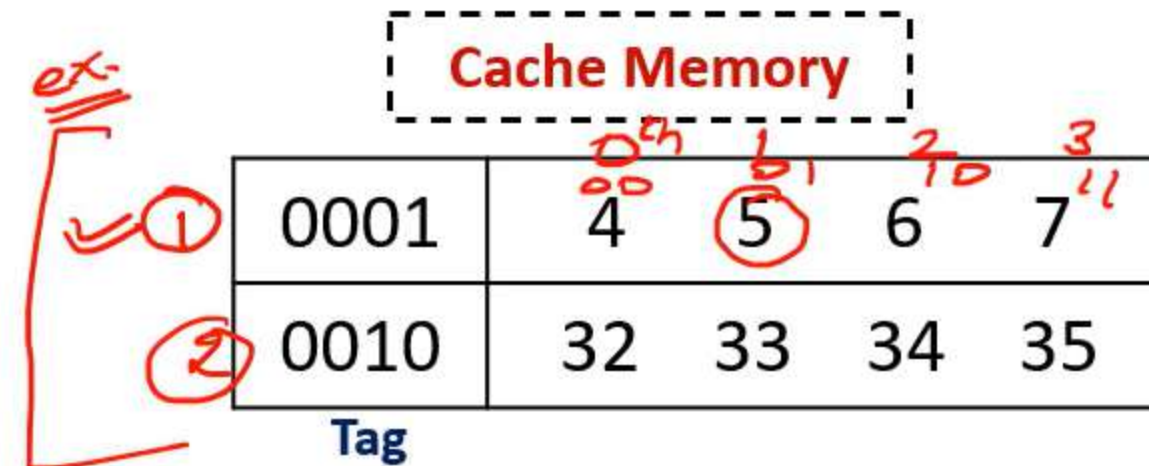
Division of Physical Address



Total words: 64

Block Size: 4

Total Blocks: 16



Example: Physical Address - 0001 01

HIT

Example: Physical Address - 0000 01

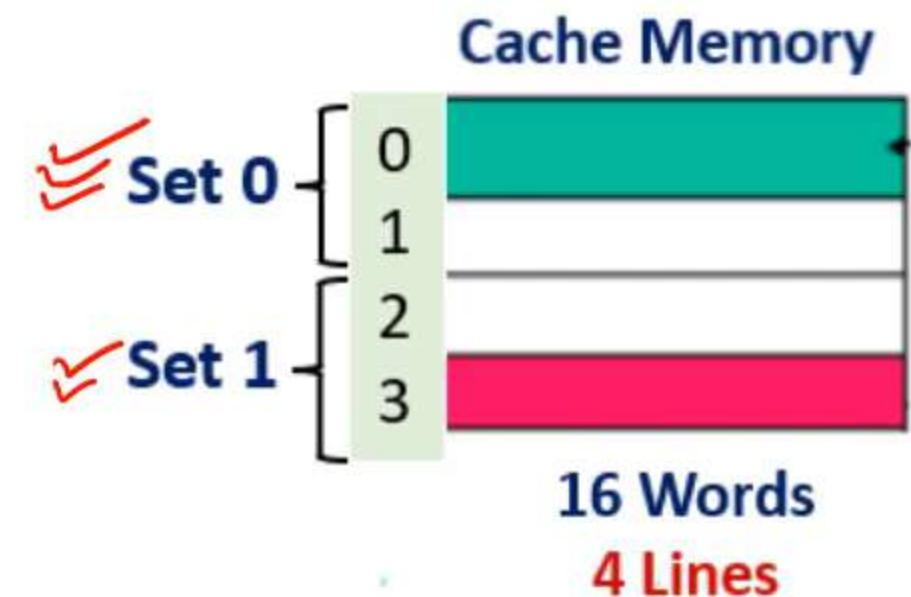
Associative Mapping Cache Organization

Set-Associative Mapping

- ✓ Set associative mapping **combines** direct mapping with associative mapping by arrangement lines of a cache into sets
- ✓ Set-Associative cache memory has **highest hit-ratio** compared to Direct Mapping and Associative Mapping Cache memory, therefore its **performance** is considerably **better**
- ✓ Set-Associative cache memory is **very expensive** because the set size increases the cost increases

- ✓ A set-associative cache that includes k lines per set is known as a **k way set-associative cache**
- ✓ Cache lines are **grouped into sets** where **each set** contains **k number of lines**
- ✓ A particular **block** of main memory can **map to only one particular set** of the cache
- ✓ However within that set, the memory block can map to any freely available cache line

2 way set-associative cache



A particular block of main memory can map to only one particular set of the cache

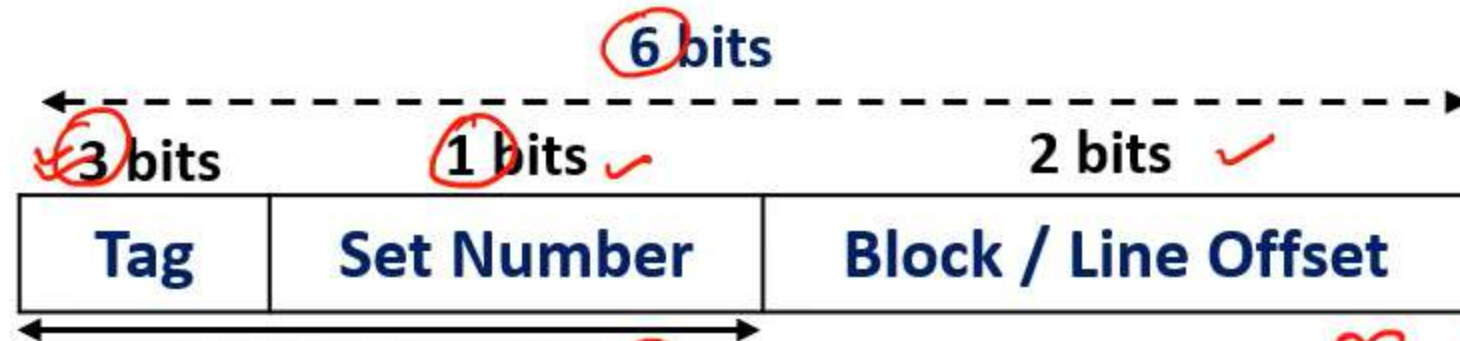
Cache Set number = (Main Memory Block Address) **Modulo** (Number of Sets in Cache)

Set-Associative Mapping

Division of Physical Address

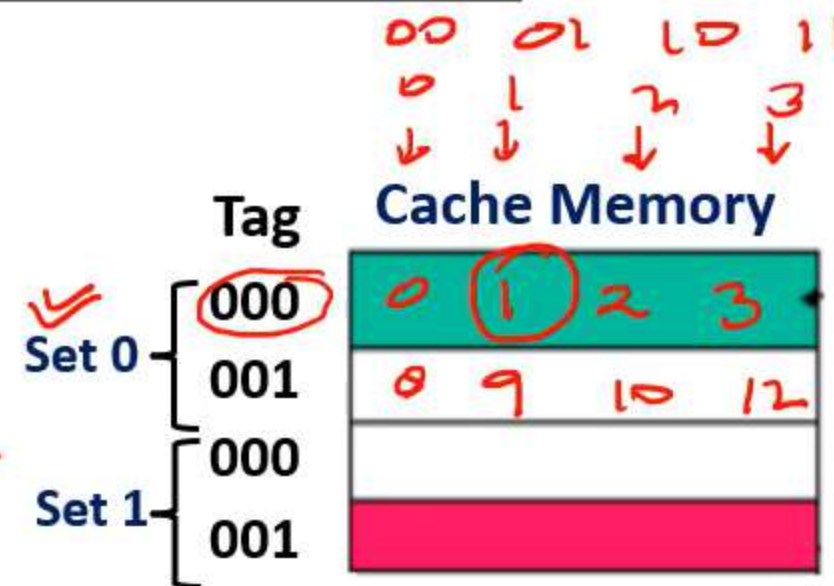
Cache Set number = (Main Memory Block Address)
Modulo (Number of Sets in Cache)

✓ $0 \bmod 2 = 0$ (Set 0) $1 \bmod 2 = 1$ (Set 1) ✓
✓ $2 \bmod 2 = 0$ (Set 0) $3 \bmod 2 = 1$ (Set 1)



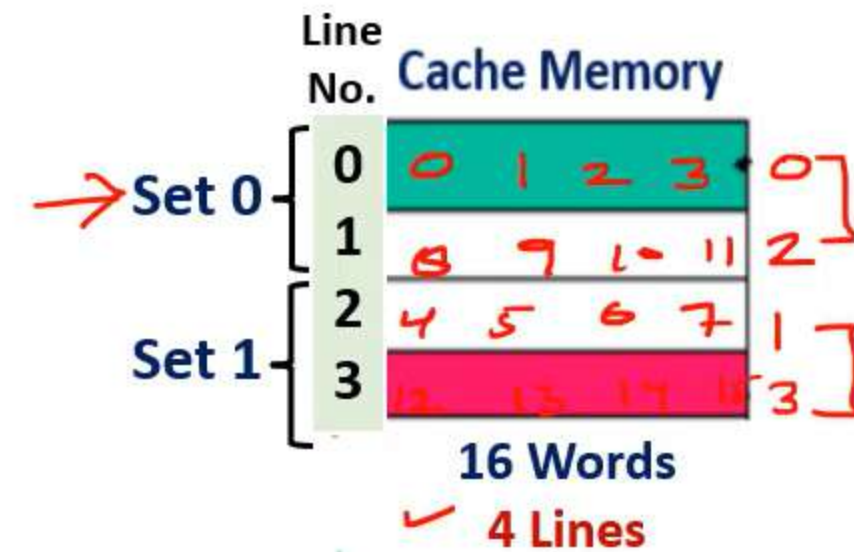
✓ Block Number (4)

Block No.	Tag	Set No.
0000	000	0
0001	000	1
0010	001	0
0011	001	1



Example: Physical Address: 000 0 01 HIT

Physical Address: 001 1 01



Total words: 64
Block Size: 4 → 2
Total Set No's: 2 → 2
Total Blocks: 16 → 2

Main Memory

0	1	2	3	0 ✓
4	5	6	7	1 ✓
8	9	10	11	2 ✓
12	13	14	15	3 ✓
16	17	18	19	4
				5
				6
				7
				8
				9
				10
				11
				12
				13
				14
				15

64 Words
16 Blocks

Set-associative Mapping Cache Organization